

[Re] Explaining in Style: Training a GAN to explain a classifier in StyleSpace

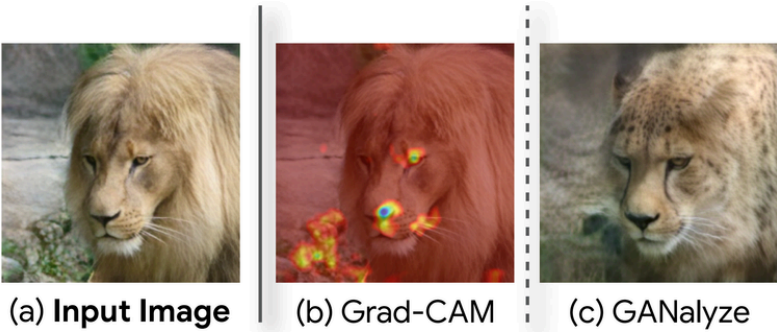
StyleEx is a novel approach for classifier-conditioned training of StyleGAN2, intending to capture classifier-specific attributes in its disentangled StyleSpace. Using the StyleEx method, the behavior of a classifier can be explained and visualized by producing counterfactual images. The original authors claim that its explanations are human-interpretable, distinct, coherent and sufficient to flip classifier predictions. We test this in our paper, by reimplementing the training code (as this was not released) and by replicating their experiments.

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01 Introduction

Classifier decisions are hard to explain: “black boxes”

- E.g. heatmaps as explanation:
 - insufficient for non-local attributes;
 - only show “where”, not “how”.

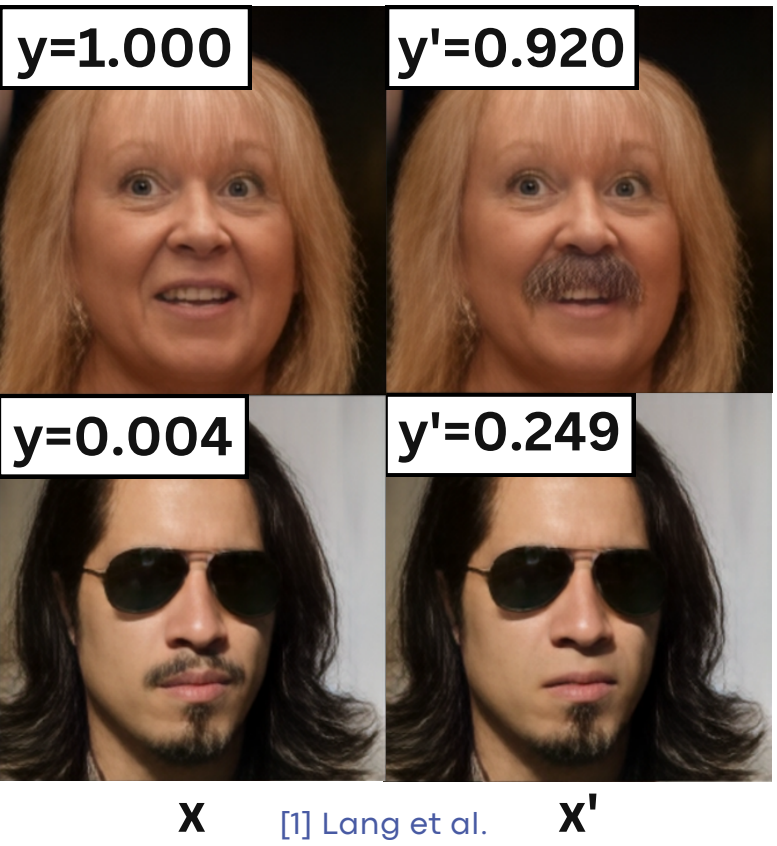


- Explaining classifiers better, can help to:
 - reveal model biases;
 - support downstream human decision-making;
 - understand our model better!

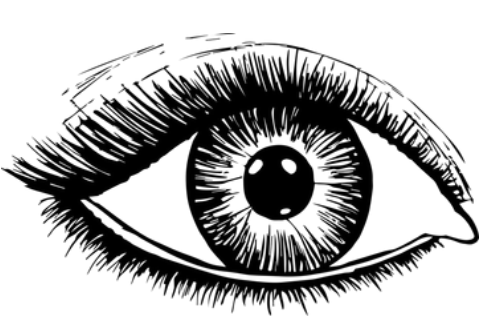
► Promising direction: counterfactual explanations

02 Counterfactuals

“Had the input x been x' , then the classifier output would have been y' instead of y ”



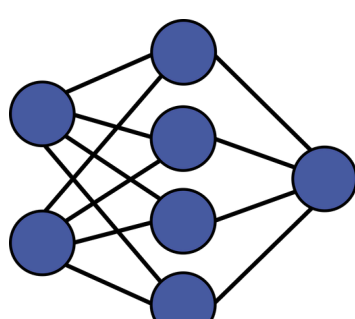
03 Scope of Reproducibility



Claim 1: Visual Coherence
Attributes detected by StyleEx should be identifiable by humans

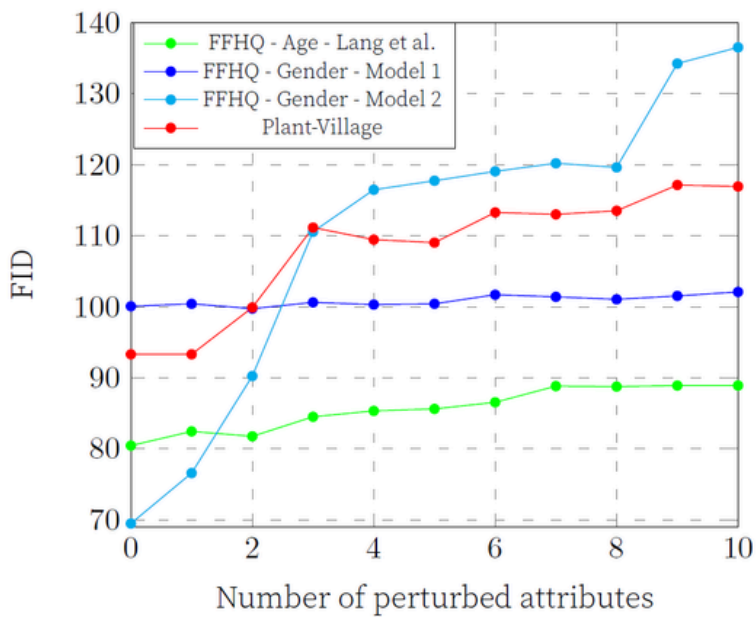
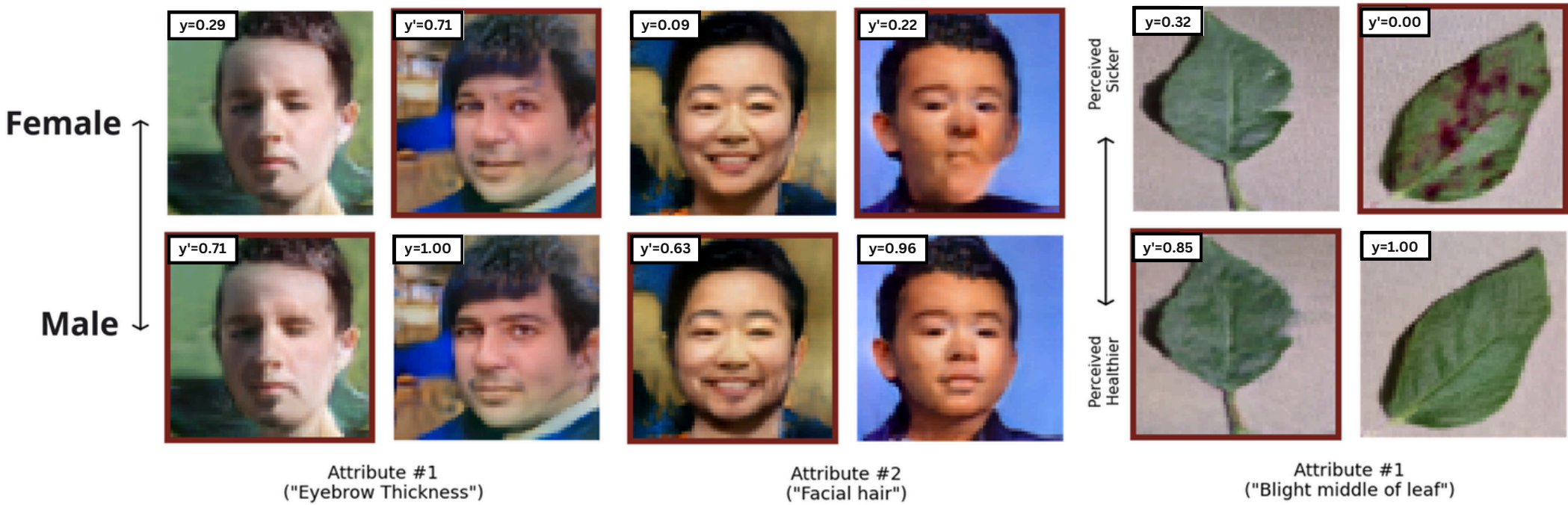


Claim 2: Distinctness
Attributes extracted by StyleEx should be distinct



Claim 3: Sufficiency
Changing attributes should result in a cumulative change of classifier output

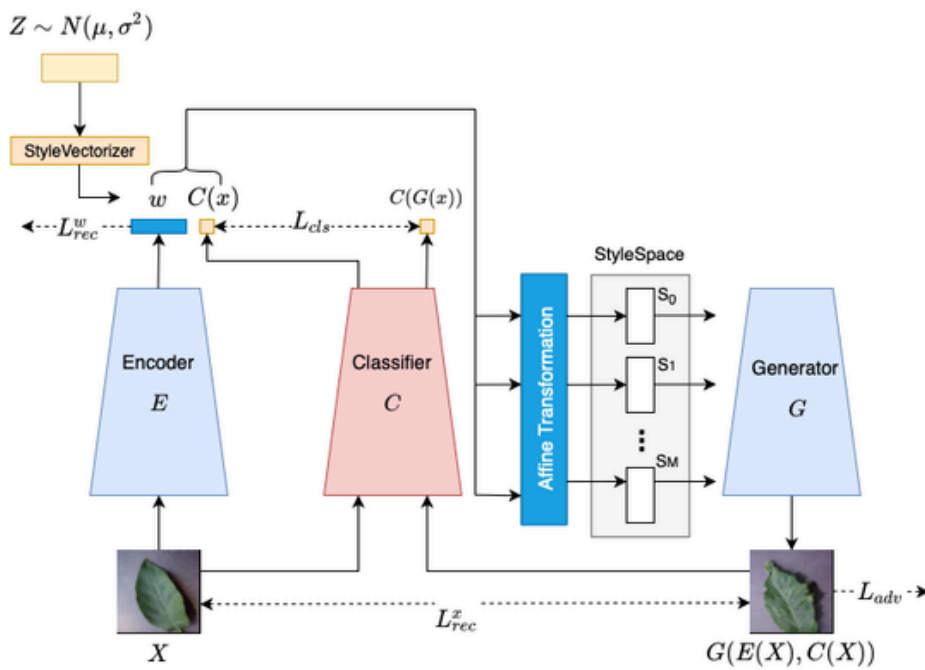
06 Results



Dataset	Ours
<i>FFHQ - Perceived Age</i>	94.8%
FFHQ - Perceived Gender (Model 1, $s = 2$)	51%
FFHQ - Perceived Gender (Model 2, $s = 1$)	21%
Plant Village - Perceived Health ($s = 2$)	30%

Table 1. Percentage of flipped classifications on different datasets. Row in italics shows our experiment on the original authors’ model. s represents the shift size used to generate the results. The shift sizes have been chosen by qualitatively looking at the produced images.

04 Architecture



05 Methodology

- Reimplemented end-to-end StyleEx training in PyTorch;
- Conducted a user study to evaluate coherence and distinctness;
- Counted classification flips to evaluate sufficiency;
- Verified sufficiency calculations on their given model.



07 Conclusion

Our experimental results support the claims posed in the original paper – the attributes detected by StyleEx are identifiable by humans to a certain degree, distinct and sufficient. However, due to the significantly lower resolution and poorer image quality of our models, these results are not directly comparable to the ones posed in the original paper.

08 Bibliography

[1] O. Lang, Y. Gandelman, M. Yarom, Y. Wald, G. Elidan, A. Hassidim, W. T. Freeman, P. Isola, A. Globerson, M. Irani, and I. Mosseri, “Explaining in style: Training a GAN to explain a classifier in StyleSpace,” 2021 IEEE/CVF International Conference on Computer Vision (ICCV), 2021.

